CHAPTER IV

DESIGN PROCESS

This chapter includes the design requirements, design implementation, as well as the testing and validation of the design that was created by the proponents for the study.

4.1 Design Requirements

The design of the automated pigpen washing system was created to provide the proponents with the guidance and basis for the project.

4.1.1 General Description of the Project

The automated pigpen washing system is a simulation-based engineering project designed to improve sanitation efficiency in small to medium scale pig farming facilities in Batangas Province, Philippines. The system is developed in response to the limitations of conventional manual pigpen cleaning practices, which are characterized by high labor demand, inconsistent cleaning performance, elevated water consumption, and occupational health risks. Through automation, the system aims to provide a more efficient, consistent, and resource-optimized cleaning process.

The proposed system consists of three (3) distinct design configurations that were developed and evaluated through mathematical modeling and computer-based simulation. These include the Initial Version of the wall-mounted sprinkler-based washing system, the Improved Version of the wall-mounted sprinkler-based washing system, and the Scraper + nozzle combination washing system. Each design represents a different approach to automated pigpen cleaning in terms of hydraulic distribution, mechanical configuration, and cleaning mechanism.

Figures 4.1, 4.2, and 4.3 present the perspective views of the three automated pigpen washing system designs, illustrating their structural configuration and operational layout as modeled in the simulation environment. Figure 4.1 shows the Initial Version of the wall-mounted sprinkler-based washing system, which represents the baseline configuration used for performance comparison. Figure 4.2 illustrates the Improved Version of the wall-mounted sprinkler-based washing system, which incorporates modifications intended to enhance cleaning efficiency and hydraulic performance. Meanwhile, Figure 4.3 presents the Scraper + nozzle combination washing system, which integrates mechanical and hydraulic cleaning mechanisms for improved waste removal capability.

Figure 4.1

Perspective View of the Initial Version of the Wall-Mounted Sprinkler-Based Washing System

Figure 4.1 presents the Initial Version of the wall-mounted sprinkler-based washing system. This design utilizes multiple wall-mounted spray nozzles positioned along the pigpen walls to distribute pressurized water across the cleaning area. The system was developed as the baseline design for evaluating the feasibility of automated pigpen washing. During the design analysis, this configuration was found to have limitations in pressure delivery, spray coverage, and overall cleaning effectiveness. Nevertheless, it served as the foundation for identifying design improvements and performance requirements for subsequent system versions.

Figure 4.2

Perspective View of the Improved Version of the Wall-Mounted Sprinkler-Based Washing System

Figure 4.2 presents the Improved Version of the wall-mounted sprinkler-based washing system. This design incorporates modifications to the nozzle arrangement, system layout, and

hydraulic distribution in order to address the limitations observed in the initial design. The improvements were intended to enhance pressure utilization, increase spray coverage uniformity, and improve overall cleaning performance. Through simulation, this design was evaluated to determine whether the implemented modifications resulted in better operational efficiency compared to the initial configuration.

Figure 4.3

Perspective View of the Scraper + Nozzle Combination Washing System

Figure 4.3 presents the Scraper + Nozzle Combination Washing System. Unlike the previous designs that rely primarily on hydraulic cleaning action, this configuration integrates a mechanical scraping mechanism with pressurized water nozzles. The scraper component assists in removing accumulated waste from the pigpen floor, while the nozzle system provides water for rinsing and cleaning. The combination of mechanical and hydraulic cleaning mechanisms is intended to improve waste removal efficiency and cleaning effectiveness, particularly in areas where water spraying alone may be insufficient. This design was evaluated and compared with the other configurations to determine its technical feasibility and overall performance under simulated operating conditions.

4.1.2. Design Standards

The proponents acknowledged the design requirements and standards that were needed for the Design, Simulation, and Analysis of an Automated Pigpen Washing System wherein these standards were given by the Philippine Electrical Code, National Electrical Manufacturers Association, and Institute of Electrical and Electronics Engineers.

4.1.2.1. Philippine Electrical Code (PEC)

Since the system operates in a wet, corrosive, and livestock-based environment, the design prioritizes personnel safety, equipment protection, and continuous operational reliability. All electrical works shall conform to PEC requirements governing wiring methods, grounding systems, utilization equipment, and protective devices to ensure safe operation under agricultural conditions.

The system shall be designed to ensure personnel safety, equipment protection, and reliable operation under wet and corrosive agricultural conditions. All electrical equipment shall be installed in accordance with approved listings and manufacturer instructions as required under PEC Article 1.10, ensuring that only approved and suitable components are used for installation in agricultural environments.

The system shall operate under low-voltage utilization not exceeding 600 volts in accordance with PEC Article 3.0. The typical configuration shall consist of a 230-volt single-phase supply or a 230/400-volt system where applicable, depending on site availability. A dedicated circuit shall be provided for the water pump motor and control system to ensure stable operation, while a separate control circuit may be used, preferably low-voltage DC, to enhance safety in automation functions. All conductors and equipment shall be properly rated for the maximum system voltage and expected load conditions to ensure safe and efficient performance.

Since the system is motor-driven through a pump or spray mechanism, its electrical design shall comply with PEC motor installation requirements. The branch circuit conductors shall have an ampacity not less than 125 percent of the motor full-load current to prevent overheating during continuous operation. The motor disconnecting means shall be rated at least 115 percent of the motor full-load current to ensure safe isolation during maintenance or fault

conditions. Motor controllers shall be selected for continuous-duty operation and shall be suitable for use in wet environments to ensure reliability during repeated washing cycles.

All wiring methods shall comply with PEC Chapter 3 provisions on wiring systems and materials. Conductors shall be installed using approved wiring methods such as conduit or moisture-resistant cable systems to ensure mechanical and environmental protection. Wiring shall be protected against mechanical damage caused by animal activity or cleaning operations, moisture ingress from high-pressure water spraying, and exposure to corrosive gases such as ammonia from pig waste. All insulation materials shall be rated for wet and damp locations, and all junction boxes shall be weatherproof or waterproof with at least IP65-level protection to ensure long-term system durability.

The grounding and bonding system shall comply with PEC Article 2.50 to ensure electrical safety in wet agricultural environments. The system shall include equipment grounding conductors connected to all metallic enclosures and motor frames, as well as bonding of all exposed non–current-carrying metal parts to eliminate potential voltage differences. A grounding electrode system shall be installed and connected to earth to provide a low-resistance fault path. Neutral-to-ground bonding shall only be performed at the service entrance to maintain a single-point grounding system. Ground resistance shall be maintained as low as practicable, preferably not exceeding five ohms, to ensure fast operation of protective devices during fault conditions. This is critical due to the high-moisture environment where the risk of electric shock is significantly elevated.

The control system, whether manual or automated using relays or PLC-based components, shall comply with PEC safety requirements for electrical control equipment. The control panel shall be fully enclosed to prevent accidental contact with energized parts, and all live components shall remain inaccessible during normal operation. An emergency stop device

shall be installed and positioned for immediate accessibility to allow rapid shutdown during abnormal conditions. Control wiring shall be physically separated from power wiring to reduce electrical noise and enhance operational safety. The control enclosure shall be designed to withstand dust, moisture, and washdown conditions typical of agricultural environments.

The system shall incorporate properly rated overcurrent protection devices in accordance with PEC requirements to safeguard electrical components from overload, short circuits, and ground faults. Circuit breakers shall be sized based on the calculated load current to ensure proper system protection. Separate protection shall be provided for motor circuits and control circuits to enhance system reliability and prevent cascading failures. These protective devices shall ensure immediate disconnection of power during motor overloads, short circuit faults, and ground faults, thereby preventing insulation damage and fire hazards.

The electrical installation shall be designed to withstand harsh agricultural environmental conditions including moisture, dust, corrosion, and chemical exposure from animal waste. Electrical components shall be installed in elevated positions or protected enclosures to avoid direct exposure to water spray during operation. Outdoor wiring systems shall utilize UV-resistant conduits and materials suitable for tropical climate conditions characterized by high humidity and frequent rainfall. All cable entry points shall be properly sealed to prevent water ingress and ensure long-term reliability of the system.

4.1.2.2. National Electrical Manufacturers Association (NEMA)

NEMA provides standardized ratings for electrical enclosures, motors, and industrial control equipment to ensure compatibility, safety, and durability in demanding applications. These standards are essential in ensuring that all electrical components used in the system can withstand exposure to water spray, ammonia vapor, dust, and other contaminants commonly present in pigpen environments.

In terms of enclosure protection, the system's electrical control panels, sensor housings, and actuator interfaces must comply with NEMA 4 or NEMA 4X enclosure ratings as defined under National Electrical Manufacturers Association guidelines. A NEMA 4-rated enclosure provides protection against splashing water and hose-directed spray, while NEMA 4X additionally provides corrosion resistance, which is particularly important in livestock environments where ammonia and organic waste gases accelerate metal degradation. As the automated washing system directly operates in a high-moisture washdown environment, the use of corrosion-resistant materials such as stainless steel or coated aluminum is required to maintain structural integrity and long-term operational reliability.

For the motor and drive system, NEMA MG 1 standards define performance characteristics for electric motors used in industrial applications. The motors driving pumps, sprayers, and mechanical scrapers in the system should be selected based on NEMA Design B or equivalent industrial duty classifications, ensuring adequate starting torque, thermal protection, and continuous duty capability. Additionally, motors should be rated for washdown duty or equipped with sealed housings to prevent water ingress during high-pressure cleaning cycles. Proper insulation classes and temperature rise limits must also be observed to avoid premature motor failure under repetitive duty cycles.

Control components such as relays, contactors, and motor starters must comply with NEMA ICS standards to ensure proper coordination and electrical safety. These components should be selected based on load category and operational duty cycle to prevent overheating and contact wear during repeated automated washing operations. Furthermore, all wiring systems must follow NEMA-recommended practices for insulation, grounding, and segregation of power and control circuits to minimize electrical noise interference and improve system reliability. Grounding systems are particularly critical in wet agricultural environments to prevent electrical hazards to both equipment and operators..

4.1.2.3. Institute of Electrical and Electronics Engineers (IEEE)

In accordance with IEEE Std 80 (Guide for Safety in AC Substation Grounding) and IEEE Std 142 (Green Book: Grounding of Industrial and Commercial Power Systems), the electrical design of the system prioritizes proper grounding, fault current mitigation, and operator protection to ensure that all metallic enclosures, pump housings, and control panels are safely bonded to earth ground. These standards are critical in preventing electric shock hazards, especially in wet and corrosive environments such as pigpens where moisture and organic waste increase conductivity risks.

For the automation and control architecture, the system follows principles aligned with IEEE standards on industrial electronics and control systems, particularly those reflected in IEEE Std 1451 (Smart Transducer Interface Standards) for sensor integration and interoperability. The use of sensors such as moisture detectors, proximity sensors, and pressure sensors is designed to ensure reliable data acquisition and standardized communication with the microcontroller or programmable logic controller (PLC). Signal conditioning, noise reduction, and analog-to-digital conversion are implemented following IEEE recommendations for measurement accuracy and electromagnetic compatibility, ensuring that sensor readings remain stable despite electrical noise from motors and pumps.

The communication and control logic of the system are designed in consideration of IEEE Std 802.15.4 for low-power wireless sensor networks (if wireless monitoring or remote control is implemented) and IEEE Std 802.11 for higher-bandwidth wireless communication in case of remote system supervision. These standards ensure that any data transmission between control nodes, monitoring interfaces, or mobile devices remains reliable, secure, and energy-efficient. For wired communication, industrial-grade serial communication protocols

(such as UART, SPI, or I2C) are implemented in accordance with IEEE-recommended practices for embedded system interoperability and deterministic data exchange.

In terms of power system design, the system adheres to IEEE guidelines for motor load management and electrical machine operation, particularly for inductive loads such as water pumps and actuators used in spraying and scraping mechanisms. Proper selection of circuit protection devices, including circuit breakers and relays, follows IEEE Std 242 (Buff Book: Protection and Coordination of Industrial Power Systems) to ensure that overloads, short circuits, and ground faults are properly isolated. This is essential to maintain continuous operation and prevent damage to critical components of the washing system.

4.1.3. Design Computation and Analysis

Below were the design computations that the proponents have conducted to achieve the desired output of the system as well as the constraints that it has. The proponents utilized different mathematical formulas to achieve these results.

Nozzle and Flow Generation Analysis

To evaluate the hydraulic performance of each proposed design, the proponents conducted a nozzle and flow generation analysis using the governing fluid mechanics equations. The computations were performed to determine the nozzle cross sectional area, volumetric flow rate, jet velocity, and Reynolds number for each design. These hydraulic parameters are essential in assessing the water discharge characteristics, spray effectiveness, and flow behavior within the washing system. The computed results also served as the basis for the succeeding analysis of pipe head loss, pressure distribution, pump power requirements, and overall system performance. The following tables present the required parameters and the corresponding computed values for each proposed design.

Design 1: Initial Wall-Mounted Sprinkler-Based Washing System

(Introductory Statement)

Table 4.1

Initial Wall-Mounted Sprinkler-Based Washing System Nozzle and Flow Generation Analysis

Parameters	Values
Nozzle Cross-Sectional Area	
Volumetric Flow Rate.	
Jet Velocity at the Nozzle Exit.	
Reynolds Numbers.	

(Table Discussion)

Design 2: Improved Wall-Mounted Sprinkler-Based Washing System

(Introductory Statement)

Table 4.2

Improved Wall-Mounted Sprinkler-Based Washing System Nozzle and Flow Generation Analysis

Parameters	Values
Nozzle Cross-Sectional Area	
Volumetric Flow Rate.	
Jet Velocity at the Nozzle Exit.	
Reynolds Numbers.	

(Table Discussion)

Design 3: Scraper + Nozzle Combination Washing System

(Introductory Statement)

Table 4.3

Scraper + Nozzle Combination Washing System Nozzle and Flow Generation Analysis

Parameters	Values
Nozzle Cross-Sectional Area	
Volumetric Flow Rate.	
Jet Velocity at the Nozzle Exit.	
Reynolds Numbers.	

(Table Discussion)

Pipe Head Loss Analysis

To evaluate the hydraulic losses within each proposed design, the proponents conducted a pipe head loss analysis using the governing principles of fluid mechanics. The computations

were performed to determine the major head loss, minor head loss, and total head loss throughout the piping system. These hydraulic parameters are essential in assessing the energy losses caused by pipe friction and flow disturbances introduced by fittings, valves, and other piping components. Determining the total head loss is necessary to evaluate the pressure available at the nozzle outlet and to verify whether the selected pump can provide sufficient pressure and flow to satisfy the operational requirements of the automated pigpen washing system. The computed results also served as the basis for the succeeding analyses of pressure distribution, pump power requirements, and overall hydraulic performance. The following tables present the required parameters and the corresponding computed values for each proposed design.

Design 1: Initial Wall-Mounted Sprinkler-Based Washing System

(Introductory Statement)

Table 4.4

Initial Wall-Mounted Sprinkler-Based Washing System Pipe Head Loss Analysis

Parameters	Values
Major Head Loss	
Minor Head Loss	
Total Head Loss.	

(Table Discussion)

Design 2: Improved Wall-Mounted Sprinkler-Based Washing System

(Introductory Statement)

Table 4.5

Improved Wall-Mounted Sprinkler-Based Washing System Pipe Head Loss Analysis

Parameters	Values

Major Head Loss	
Minor Head Loss	
Total Head Loss.	

(Table Discussion)

Design 3: Scraper + Nozzle Combination Washing System

(Introductory Statement)

Table 4.6

Scraper + Nozzle Combination Washing System Pipe Head Loss Analysis

Parameters	Values
Major Head Loss	
Minor Head Loss	
Total Head Loss.	

(Table Discussion)

Pressure and Energy Balance Analysis

To evaluate the hydraulic pressure distribution and energy characteristics of each proposed design, the proponents conducted a pressure and energy balance analysis using Bernoulli's equation with head losses. The computations were performed to determine the relationship between the inlet and outlet pressures while accounting for velocity changes, elevation differences, and the total head loss throughout the piping system. This analysis is essential in assessing whether sufficient pressure is maintained at the nozzle outlet to achieve effective cleaning performance and to verify that the selected pump can overcome the hydraulic resistance of the system. The computed results also served as the basis for evaluating the overall hydraulic performance and operational feasibility of each proposed design. The following

tables present the required parameters and the corresponding computed values for each proposed design.

Design 1: Initial Wall-Mounted Sprinkler-Based Washing System

(Introductory Statement)

Table 4.7

Initial Wall-Mounted Sprinkler-Based Washing System Pressure and Energy Balance Analysis

Parameters	Values
Bernoulli's Equation with Head Loss	

(Table Discussion)

Design 2: Improved Wall-Mounted Sprinkler-Based Washing System

(Introductory Statement)

Table 4.8

Improved Wall-Mounted Sprinkler-Based Washing System Pressure and Energy Balance Analysis

Parameters	Values
Bernoulli's Equation with Head Loss	

(Table Discussion)

Design 3: Scraper + Nozzle Combination Washing System

(Introductory Statement)

Table 4.9

Scraper + Nozzle Combination Washing System Pressure and Energy Balance Analysis

Parameters	Values
Bernoulli's Equation with Head Loss	

(Table Discussion)

Pump Power Requirement Analysis

To determine the power requirements of each proposed design, the proponents conducted a pump power requirement analysis using the governing hydraulic power equations. The computations were performed to determine the hydraulic power and pump input power required to deliver the desired flow rate while overcoming the total system head. These parameters are essential in evaluating whether the selected pump can provide sufficient power to operate the washing system efficiently under the calculated hydraulic conditions. Furthermore, the computed pump power requirements served as the basis for assessing the energy demand and operational feasibility of each proposed design. The following tables present the required parameters and the corresponding computed values for each proposed design.

Design 1: Initial Wall-Mounted Sprinkler-Based Washing System

(Introductory Statement)

Table 4.10

Initial Wall-Mounted Sprinkler-Based Washing System Pump Power Requirement Analysis

Parameters	Values
Hydraulic Power	
Pump Input Power	

(Table Discussion)

Design 2: Improved Wall-Mounted Sprinkler-Based Washing System

(Introductory Statement)

Table 4.11

Improved Wall-Mounted Sprinkler-Based Washing System Pump Power Requirement Analysis

Parameters	Values

Hydraulic Power	
Pump Input Power	

(Table Discussion)

Design 3: Scraper + Nozzle Combination Washing System

(Introductory Statement)

Table 4.12

Scraper + Nozzle Combination Washing System Pump Power Requirement Analysis

Parameters	Values
Hydraulic Power	
Pump Input Power	

(Table Discussion)

Mechanical Scraper System Analysis

To evaluate the mechanical performance of the scraper mechanism incorporated in the proposed design, the proponents conducted a mechanical scraper system analysis using the governing principles of engineering mechanics. The computations were performed to determine the frictional force, required torque, angular velocity, and mechanical power of the scraper drive system. These mechanical parameters are essential in assessing the resistance encountered during scraper movement, the rotational force required to drive the mechanism, the operating speed of the motor, and the corresponding power demand of the system. The computed results served as the basis for verifying the adequacy of the selected motor and ensuring the reliable operation of the scraper mechanism under the expected loading conditions. The following tables present the required parameters and the corresponding computed values for the proposed scraper based washing system.

Design 1: Initial Wall-Mounted Sprinkler-Based Washing System

(Introductory Statement)

Table 4.13

Initial Wall-Mounted Sprinkler-Based Washing System Mechanical Scraper System Analysis

Parameters	Values
Frictional Force	
Torque	
Angular velocity	
Mechanical Power	

(Table Discussion)

Design 2: Improved Wall-Mounted Sprinkler-Based Washing System

(Introductory Statement)

Table 4.14

Improved Wall-Mounted Sprinkler-Based Washing System Mechanical Scraper System Analysis

Parameters	Values
Frictional Force	
Torque	
Angular velocity	
Mechanical Power	

(Table Discussion)

Design 3: Scraper + Nozzle Combination Washing System

(Introductory Statement)

Table 4.15

Scraper + Nozzle Combination Washing System Mechanical Scraper System Analysis

Parameters	Values
Frictional Force	
Torque	
Angular velocity	
Mechanical Power	

(Table Discussion)

Simulation Validation and Accuracy Analysis

To evaluate the reliability of the developed MATLAB and Simulink models, the proponents conducted a simulation validation and accuracy analysis comparing the simulated results with the corresponding theoretical computations. The validation process involved determining the percentage error between the theoretical and simulated values of the selected hydraulic and mechanical parameters. This analysis verifies the accuracy of the simulation model in representing the actual behavior of the proposed Automated Pigpen Washing System and assesses its suitability for predicting system performance under the specified operating conditions. The following table presents the theoretical values, simulated results, percentage error, and corresponding interpretation for each proposed design.

Design 1: Initial Wall-Mounted Sprinkler-Based Washing System

(Introductory Statement)

Table 4.16

Initial Wall-Mounted Sprinkler-Based Washing System Simulation Validation and Accuracy Analysis

Parameters	Values
Percentage Error	

(Table Discussion)

Design 2: Improved Wall-Mounted Sprinkler-Based Washing System

(Introductory Statement)

Table 4.17

Improved Wall-Mounted Sprinkler-Base Washing System Simulation Validation and Accuracy Analysis

Parameters	Values
Percentage Error	

(Table Discussion)

Design 3: Scraper + Nozzle Combination Washing System

(Introductory Statement)

Table 4.18

Scraper + Nozzle Combination Washing System Simulation Validation and Accuracy Analysis

Parameters	Values
Percentage Error	

(Table Discussion)

Water and Energy Consumption Analysis

To evaluate the operational efficiency of each proposed design, the proponents conducted a water and energy consumption analysis using the governing equations. The computations were performed to determine the total water consumption, electrical energy consumption, and specific water energy consumption of each design during a single cleaning cycle. These performance parameters are essential in assessing the efficiency, sustainability, and operating cost of the automated pigpen washing system. The computed results were used to compare the resource utilization of the three proposed designs and to identify the configuration that provides effective cleaning performance while minimizing water and electrical

energy consumption. The following tables present the required parameters and the corresponding computed values for each proposed design.

Design 1: Initial Wall-Mounted Sprinkler-Based Washing System

(Introductory Statement)

Table 4.19

Initial Wall-Mounted Sprinkler-Based Washing System Water and Energy Consumption Analysis

Parameters	Values
Total Water Consumption	
Electrical Energy Consumption	
Specific Water–Energy Consumption	

(Table Discussion)

Design 2: Improved Wall-Mounted Sprinkler-Based Washing System

(Introductory Statement)

Table 4.20

Improved Wall-Mounted Sprinkler-Based Washing System Water and Energy Consumption Analysis

Parameters	Values
Total Water Consumption	
Electrical Energy Consumption	
Specific Water–Energy Consumption	

(Table Discussion)

Design 3: Scraper + Nozzle Combination Washing System

(Introductory Statement)

Table 4.21

Scraper + Nozzle Combination Washing System Water and Energy Consumption Analysis

Parameters	Values
Total Water Consumption	
Electrical Energy Consumption	
Specific Water–Energy Consumption	

(Table Discussion)

4.1.4 Materials and Components

The materials and components considered in this study are categorized into three major classifications: mechanical and structural components, electrical components, and washing system components. These categories encompass the essential elements required for the design, simulation, and analysis of the proposed automated pigpen washing system. Each component is identified and organized according to its respective function within the system to establish a clear reference for the proponents during the design process. The corresponding quantity and specifications of the materials and components are presented in the bill of materials to facilitate systematic evaluation, documentation, and technical analysis of the proposed system design.

Table 4.22

Table of Materials and Components for the Three Designs

Materials and Components	**Design 1 Initial Wall-Mounted Sprinkler-Based Washing System**		**Design 2 Improved Wall-Mounted Sprinkler-Based Washing System**		**Design 3 Scraper + Nozzle Combination Washing System**	
	Quantity	Amount	Quantity	Amount	Quantity	Amount

(Table Discussion)

4.1.5. Design Layout

The design layouts of the proposed automated pigpen washing system configurations were developed to present the overall system arrangement, component placement, and operational structure of each design alternative. The layouts serve as a visual representation of the proposed designs, allowing the evaluation of the system architecture, mechanical arrangement, hydraulic components, and electrical control integration. These design layouts were prepared to provide a clearer understanding of the proposed washing system configurations and their respective operational mechanisms.

Design 1: Design Layout of the Wall Mounted Sprinkler Washing System

(Figure)

The first design layout presents the configuration of the wall mounted sprinkler washing system. The layout illustrates the arrangement of the major system components, including the water supply line, piping network, spray nozzles, pump assembly, control components, and electrical connections required for system operation.

The design layout includes the dimensional representation of the pigpen area, nozzle placement, piping arrangement, and the positioning of electrical and mechanical components. The layout also presents the perspective view of the proposed system to demonstrate the actual positioning and coverage area of the sprinkler mechanism. Furthermore, the design includes the electrical schematic, system specifications, and operational details that define the integration of the hydraulic and electrical components within the automated washing system.

Design 2: Design Layout of the Improved Wall Mounted Sprinkler Washing System

(Figure)

The second design layout presents the improved configuration of the wall mounted sprinkler washing system. This design incorporates modifications in the arrangement and positioning of the washing components to enhance cleaning coverage, water distribution, and overall system efficiency.

The layout illustrates the updated placement of the spray nozzles, piping system, water control components, and electrical connections. Similar to the first design, the layout includes the floor plan with dimensions, component specifications, hydraulic arrangement, electrical system design, and perspective views of the proposed configuration. These elements provide a detailed representation of the design improvements implemented to optimize the washing performance of the system under simulated operating conditions.

Design 3: Design Layout of the Scraper and Nozzle Combination Washing System

(Figure)

The third design layout presents the configuration of the scraper and nozzle combination washing system. This design integrates a mechanical scraping mechanism with a pressurized nozzle system to improve the removal of waste materials and enhance the cleaning process within the pigpen area.

The design layout illustrates the arrangement of the linear scraper mechanism, nozzle locations, water supply system, drive components, and electrical control system. It also includes the dimensional layout of the pigpen, mechanical assembly details, electrical connections, system specifications, and perspective views of the proposed design. These components provide a comprehensive representation of the integration between the mechanical and hydraulic systems, which are evaluated through simulation and engineering analysis.

The following table presents the similarities and differences among the three proposed automated pigpen washing system designs developed by the proponents. The similarities include the utilization of automated control mechanisms, water delivery components, electrical power integration, and the application of engineering design principles. Meanwhile, the differences are observed in the cleaning mechanisms, hydraulic configurations, component arrangements, and expected performance parameters such as cleaning coverage, water consumption, and energy requirements. These variations serve as the basis for the comparative analysis and simulation evaluation of each proposed design.

Table 4.23 (Landscape)

Similarities and Dissimilarities Between the Designs

Design	Similarities	Dissimilarities
(Figure) Design 1: Wall Mounted Sprinkler Washing System	•	•
(Figure) Design 2: Improved Wall Mounted Sprinkler Washing System		
(Figure) Design 3: Scraper and Nozzle Combination Washing System		

(Table Discussion)

4.1.6 Circuit Diagram

The circuit diagrams of the proposed automated pigpen washing system designs illustrate the electrical connections and control configurations required for system operation. These diagrams provide a technical representation of the electrical components, power supply connections, switching devices, and control circuits integrated into each design. They also establish the relationship between the electrical and mechanical components, particularly the water pump, automated control system, and other electrical devices responsible for the washing process. The circuit diagrams serve as a reference for evaluating the electrical configuration, operational sequence, and system integration of the proposed designs.

Circuit Diagram of the Proposed Designs

The following circuit diagrams present the electrical configurations of the three proposed automated pigpen washing system designs developed by the proponents. Each diagram illustrates the arrangement and interconnection of the electrical components required to operate the washing mechanism. These diagrams also provide a basis for analyzing the power distribution, control logic, and electrical protection requirements of each design.

Design 1: Circuit Diagram of the Wall Mounted Sprinkler Washing System

The circuit diagram of the first design presents the electrical connections of the wall mounted sprinkler washing system. It illustrates the integration of the power supply, protective devices, switching components, and control circuit responsible for operating the water pump and sprinkler mechanism. The diagram also identifies the connections required to activate the washing system and regulate its operation through the designated control components. This configuration serves as a reference for evaluating the electrical functionality and operational requirements of the first proposed design.

(Figure)

(Discussion)

Design 2: Circuit Diagram of the Improved Wall Mounted Sprinkler Washing System

The circuit diagram of the second design illustrates the electrical configuration of the improved wall mounted sprinkler washing system. It presents the arrangement of the power supply, electrical protection devices, switching components, and control circuit used to operate the washing mechanism. The diagram also shows the electrical connections associated with the modified system configuration and its control components. These connections provide the basis for evaluating the electrical integration, control functionality, and operational performance of the improved design.

(Figure)

(Discussion)

Design 3: Circuit Diagram of the Scraper and Nozzle Combination Washing System

The circuit diagram of the third design presents the electrical configuration of the scraper and nozzle combination washing system. It illustrates the integration of the power supply, protective devices, motor control components, and electrical connections required to operate the mechanical scraper and water spraying mechanism. The diagram also identifies the control connections that coordinate the operation of the scraping mechanism and the water delivery system. This configuration provides a basis for examining the electrical coordination between the mechanical and hydraulic components and evaluating the operational requirements of the third proposed design.

(Figure)

(Discussion)

The circuit diagrams of the three designs support the simulation and comparative analysis of the proposed automated pigpen washing system. Through these diagrams, the proponents can examine the electrical interconnections, control functions, and compatibility of the system components. The resulting analysis provides a technical basis for assessing the operational feasibility, electrical safety, and overall performance of each design alternative.

4.1.7 Bill of Materials and Specification

The bill of materials presents the components, materials, quantities, and estimated costs required for the fabrication and implementation of the three proposed automated pigpen washing system designs. The proponents organized the materials according to their respective functions within the system, including mechanical components, hydraulic components, electrical devices, and control system components. The corresponding tables specify the quantity and unit cost of each item, together with the total estimated expenditure in Philippine pesos. This information provides a basis for evaluating the material requirements, estimating the overall project cost, and comparing the economic feasibility of the proposed designs.

Bill of Materials and Components of the Proposed Designs

The following tables present the electrical components and materials required for each proposed automated pigpen washing system design. These components support the power distribution, motor operation, automated control, and electrical protection functions of the system. Each table identifies the required materials, quantities, unit prices, and total costs to establish the estimated electrical expenses for each design alternative.

Design 1: Bill of Materials and Components for the Wall Mounted Sprinkler Washing System

Table X

Description	Quantity	Unit	Unit Price	Total Price
Fabrication	-	-	-	PHP 60,000.00
Arduino Mega	1	piece	PHP 600.00	PHP 600.00
Axial Flow fan 6x6 12V	2	piece	PHP 100.00	PHP 200.00

DC 12V 15A Power Supply Adapter Converter	1	piece	PHP 234.00	PHP 702.00
Relay Module with Optocoupler	5	piece	PHP 127.00	PHP 630.00
Pressure Transducer Transmitter Sensor	1	piece	P800.00	PHP 800.00
NEMA 17 Stepper Motor	2	piece	P900.00	PHP 1,800.00
NEMA 23 Stepper Motor	3	piece	P2,500.00	PHP 7,500.00
Love Joint 8 mm x 8 mm	3	piece	P170.00	PHP 510.00
PCB Board Green Double Sided Protoboard P66	2	piece	PHP 350.00	PHP 700.00
Weatherproof Enclosure 285x189x70mm	1	piece	PHP 1,200.00	PHP 1,200.00
WP Enclosure Plastic Junction Box 83x58x33mm	1	piece	PHP 150.00	PHP 150.00
Acrylic Board 3mm	1	piece	PHP 250.00	PHP 250.00
Rocker Switch	1	piece	PHP 125.00	PHP 125.00
Imaging Sensor	1	piece	PHP 1000.00	PHP 1,000.00
Circuit Breaker	2	piece	PHP 489.00	PHP 1,038.00
Electrical Tape	3	piece	PHP 48.00	PHP 144.00
THWN 3.5 Mm^2	2	meter	PHP 519.00	PHP 1,038.00

Spray Nozzle	21	piece	PHP 79.00	PHP 1,975.00
High-Pressure Power Hose	10	meter	PHP 40.00	PHP 390.00
Brass Solenoid Valve 1/4"	5	piece	PHP 420.00	PHP 420.00
1.5 Hp High-Pressure Water Pump Set	1	piece	PHP 8,000.00	PHP 8,000.00
Brass Coupler Adapter Reducer	15	piece	PHP 35.00	PHP 525.00
PPR Pipes	10	meter	PHP 87.00	PHP 870.00
PPR Fittings	40	piece	PHP 108.00	PHP 4,320.00
			Materials And Components Subtotal:	**PHP 94,887.00**

(Table Discussion)

Design 2: Bill of Materials and Components for the Improved Wall Mounted Sprinkler Washing System

Table X

Description	Quantity	Unit	Unit Price	Total Price
Fabrication	-	-	-	PHP 60,000.00
Arduino Mega	1	piece	PHP 600.00	PHP 600.00
Axial Flow fan 6x6 12V	2	piece	PHP 100.00	PHP 200.00

DC 12V 15A Power Supply Adapter Converter	1	piece	PHP 234.00	PHP 702.00
Relay Module with Optocoupler	5	piece	PHP 127.00	PHP 630.00
Pressure Transducer Transmitter Sensor	1	piece	P800.00	PHP 800.00
NEMA 17 Stepper Motor	2	piece	P900.00	PHP 1,800.00
NEMA 23 Stepper Motor	3	piece	P2,500.00	PHP 7,500.00
Love Joint 8 mm x 8 mm	3	piece	P170.00	PHP 510.00
PCB Board Green Double Sided Protoboard P66	2	piece	PHP 350.00	PHP 700.00
Weatherproof Enclosure 285x189x70mm	1	piece	PHP 1,200.00	PHP 1,200.00
WP Enclosure Plastic Junction Box 83x58x33mm	1	piece	PHP 150.00	PHP 150.00
Acrylic Board 3mm	1	piece	PHP 250.00	PHP 250.00
Rocker Switch	1	piece	PHP 125.00	PHP 125.00
Imaging Sensor	1	piece	PHP 1000.00	PHP 1,000.00
Circuit Breaker	2	piece	PHP 489.00	PHP 1,038.00
Electrical Tape	3	piece	PHP 48.00	PHP 144.00
THWN 3.5 Mm^2	2	meter	PHP 519.00	PHP 1,038.00

Spray Nozzle	21	piece	PHP 79.00	PHP 1,975.00
High-Pressure Power Hose	10	meter	PHP 40.00	PHP 390.00
Brass Solenoid Valve 1/4"	5	piece	PHP 420.00	PHP 420.00
1.5 Hp High-Pressure Water Pump Set	1	piece	PHP 8,000.00	PHP 8,000.00
Brass Coupler Adapter Reducer	15	piece	PHP 35.00	PHP 525.00
PPR Pipes	10	meter	PHP 87.00	PHP 870.00
PPR Fittings	40	piece	PHP 108.00	PHP 4,320.00
			Materials And Components Subtotal:	**PHP 94,887.00**

(Table Discussion)

Design 3: Bill of Materials and Components for the Scraper and Nozzle Combination Washing System

Table X

Description	Quantity	Unit	Unit Price	Total Price
Fabrication	-	-	-	PHP 60,000.00
Arduino Mega	1	piece	PHP 600.00	PHP 600.00
Axial Flow fan 6x6 12V	2	piece	PHP 100.00	PHP 200.00

DC 12V 15A Power Supply Adapter Converter	1	piece	PHP 234.00	PHP 702.00
Relay Module with Optocoupler	5	piece	PHP 127.00	PHP 630.00
Pressure Transducer Transmitter Sensor	1	piece	P800.00	PHP 800.00
NEMA 17 Stepper Motor	2	piece	P900.00	PHP 1,800.00
NEMA 23 Stepper Motor	3	piece	P2,500.00	PHP 7,500.00
Love Joint 8 mm x 8 mm	3	piece	P170.00	PHP 510.00
PCB Board Green Double Sided Protoboard P66	2	piece	PHP 350.00	PHP 700.00
Weatherproof Enclosure 285x189x70mm	1	piece	PHP 1,200.00	PHP 1,200.00
WP Enclosure Plastic Junction Box 83x58x33mm	1	piece	PHP 150.00	PHP 150.00
Acrylic Board 3mm	1	piece	PHP 250.00	PHP 250.00
Rocker Switch	1	piece	PHP 125.00	PHP 125.00
Imaging Sensor	1	piece	PHP 1000.00	PHP 1,000.00
Circuit Breaker	2	piece	PHP 489.00	PHP 1,038.00
Electrical Tape	3	piece	PHP 48.00	PHP 144.00
THWN 3.5 Mm^2	2	meter	PHP 519.00	PHP 1,038.00

Spray Nozzle	21	piece	PHP 79.00	PHP 1,975.00
High-Pressure Power Hose	10	meter	PHP 40.00	PHP 390.00
Brass Solenoid Valve 1/4"	5	piece	PHP 420.00	PHP 420.00
1.5 Hp High-Pressure Water Pump Set	1	piece	PHP 8,000.00	PHP 8,000.00
Brass Coupler Adapter Reducer	15	piece	PHP 35.00	PHP 525.00
PPR Pipes	10	meter	PHP 87.00	PHP 870.00
PPR Fittings	40	piece	PHP 108.00	PHP 4,320.00
			Materials And Components Subtotal:	**PHP 94,887.00**

(Table Discussion)

Multiple Design and Constraints. After determining the configurations, operational requirements, component specifications, and estimated material costs of the three proposed automated pigpen washing system designs, the proponents identified the principal design constraints associated with each alternative. These constraints include the cleaning mechanism, water consumption, energy requirements, cleaning coverage, operating time, component availability, and estimated implementation cost. The identified parameters provide a basis for comparing the technical performance and economic feasibility of each proposed design.

Table X

Multiple Design and Multiple Constraints

Constraints	Design 1	Design 2	Design 3

The comparative table presents the characteristics and limitations of the three proposed designs. Design 1 employs a wall mounted sprinkler mechanism that distributes water through strategically positioned nozzles to clean the pigpen area. Design 2 incorporates improvements to the wall mounted sprinkler configuration to address potential limitations in water distribution and cleaning coverage. Design 3 combines a mechanical scraping mechanism with a nozzle based washing system to facilitate the removal of solid waste and improve the overall cleaning process.

The designs differ in their mechanical configurations, hydraulic requirements, electrical power consumption, and expected cleaning performance. The first and second designs primarily

depend on water distribution through sprinkler nozzles, whereas the third design integrates mechanical scraping with water spraying. Consequently, the proponents must evaluate each alternative in terms of cleaning effectiveness, water consumption, energy requirements, operating time, and system complexity.

The estimated total cost also serves as an important constraint in selecting the most suitable design. The cost comparison considers the materials and components required for each system, including the mechanical assembly, water delivery components, electrical devices, and automated control equipment. By examining these factors alongside the results of the design simulations and engineering analyses, the proponents can determine which configuration offers the most appropriate balance between cleaning performance, resource efficiency, operational reliability, and implementation cost.

4.2. Design Implementation

The proponents implemented the proposed automated pigpen washing system designs through computer aided modeling, simulation, and engineering analysis to evaluate their respective operational characteristics and performance. The process involved developing the system models, defining the relevant design parameters, and analyzing the simulation results to determine the feasibility and effectiveness of each proposed configuration. The evaluation focused on key performance indicators, including water consumption, water flow rate, nozzle discharge velocity, cleaning coverage, electrical power consumption, operating time, and estimated implementation cost. These parameters provided a quantitative basis for comparing the three proposed designs and identifying their respective strengths and limitations.

The proponents utilized MATLAB Simulink to develop and simulate the mathematical models representing the hydraulic, mechanical, and electrical characteristics of the proposed automated pigpen washing systems. The simulation results were evaluated against theoretical calculations derived from established engineering equations and design specifications. This comparative approach enabled the proponents to assess the consistency of the simulation outputs with the expected system behavior and determine the suitability of each design under the specified operating conditions.

4.2.1. MATLAB Simulink Model Development

The MATLAB Simulink model served as the primary computational framework for simulating and analyzing the proposed automated pigpen washing system designs. The model incorporated the relevant mathematical relationships and engineering parameters required to evaluate the hydraulic performance, electrical requirements, and operational characteristics of each configuration. The simulation outputs provided the quantitative data necessary for assessing system performance and conducting comparative analysis.

(Figure)

(Entire MATLAB simulink model)

The model consisted of interconnected subsystems developed to calculate the parameters relevant to the study. These subsystems represented the principal operational aspects of the washing system, including water flow characteristics, nozzle performance, water consumption, and electrical energy requirements. Depending on the configuration, the model also incorporated parameters associated with the operation of the mechanical scraping

mechanism. Each subsystem performed specific calculations and transferred its outputs to other components of the model to support the overall simulation process.

The proponents constructed the computational model using fundamental Simulink blocks, including Constant, Product, Divide, and Sum blocks, to implement the mathematical equations governing the system. The Constant blocks allowed the proponents to modify input parameters, such as nozzle diameter, water pressure, flow rate, operating duration, pump power, and other relevant design specifications. This configuration facilitated the evaluation of different operating conditions and enabled consistent comparisons among the three proposed designs.

To simplify the monitoring and interpretation of simulation results, the proponents utilized Outport and related signal-routing blocks to transfer calculated outputs from individual subsystems to the main model interface. This arrangement consolidated the essential performance indicators and reduced the need to inspect each subsystem separately. The proponents organized the principal outputs within a designated blue-colored area to improve the visibility and accessibility of the simulation results.

The principal outputs included the calculated volumetric flow rate, nozzle discharge velocity, water consumption, estimated electrical power demand, energy consumption, and other applicable performance indicators. The specific parameters depended on the configuration and operational requirements of each proposed design. The proponents subsequently compared these outputs with the corresponding theoretical calculations to evaluate the accuracy and consistency of the simulation model. The resulting data served as the basis for the comparative assessment of the three automated pigpen washing system designs in terms of hydraulic performance, electrical energy requirements, resource utilization, and overall operational feasibility.

CHAPTER V

RESULTS AND DISCUSSION

This chapter presents the results and discussion of the study entitled *Design, Simulation, and Analysis of an Automated Pigpen Washing System*. It discusses the findings obtained from the design development, mathematical modeling, computer simulation, and comparative analysis of the three proposed system configurations. The chapter also presents the summary of findings, conclusions derived from the results, and recommendations for further development, testing, and improvement of the proposed automated washing system.

5.1. Discussion of Results

The following section presents the results obtained from the design, simulation, and analysis of the three proposed automated pigpen washing system configurations. The discussion focuses on their structural and mechanical arrangements, hydraulic performance, electrical requirements, estimated material costs, and operational characteristics. The proponents evaluated these parameters to determine the strengths, limitations, and overall feasibility of each design.

1. Design Development and Engineering Analysis

The study developed three proposed configurations for an automated pigpen washing system to improve the efficiency and consistency of pigpen cleaning. The designs incorporated different cleaning mechanisms to address the removal of accumulated waste materials and the distribution of water across the pigpen area.

Design 1 employed a wall mounted sprinkler washing mechanism that utilized strategically positioned spray nozzles to distribute water across the designated cleaning area.

This configuration focused on automating the washing process through a water delivery system controlled by the associated electrical components. Design 2 introduced improvements to the wall mounted sprinkler configuration to enhance water distribution and cleaning coverage. Design 3 combined a mechanical scraping mechanism with a nozzle based washing system to facilitate the removal of solid waste and support subsequent washing operations.

The proponents developed the design layouts, circuit diagrams, and component arrangements to establish the technical requirements of each configuration. The designs incorporated the necessary mechanical, hydraulic, and electrical components to support automated operation. The electrical system provided the power and control functions required to operate the washing mechanisms, while the hydraulic system supplied and distributed water through the designated piping and nozzle arrangements. For the third design, the mechanical scraping mechanism provided an additional means of removing accumulated solid waste.

The proponents considered relevant engineering principles and applicable safety requirements during the design process. The electrical configuration required appropriate conductors, overcurrent protection devices, switching components, and control devices based on the system specifications. The hydraulic configuration required suitable piping, nozzle dimensions, water pressure, and flow characteristics to achieve the intended cleaning performance. These considerations established the engineering basis for evaluating the feasibility and operational requirements of the proposed designs.

To determine the hydraulic requirements of each configuration, the proponents considered the governing fluid mechanics equations and the specified design parameters. These included nozzle cross sectional area, volumetric flow rate, discharge velocity, and Reynolds number, where applicable. The calculated parameters provided information about the water discharge characteristics and flow behavior of the washing system. The proponents also

considered water consumption and operating duration to estimate the water requirements associated with the cleaning process.

The electrical analysis focused on the power requirements of the components used in each design. The proponents considered the electrical ratings of the water pump, drive motor, control devices, and other electrical loads applicable to each configuration. These parameters provided a basis for estimating the electrical energy consumption during operation and evaluating the compatibility of the electrical components with the intended system requirements.

The proponents also prepared the bill of materials for each design to identify the required components, quantities, unit prices, and estimated total costs. The cost analysis considered the mechanical assemblies, hydraulic components, electrical devices, and control equipment required for implementation. This information supported the comparison of the estimated implementation costs and the identification of potential cost constraints associated with each proposed configuration.

2. Simulation Results and Comparison of Theoretical and Simulated Values

The proponents utilized MATLAB Simulink to model and simulate the relevant operational characteristics of the three proposed automated pigpen washing system designs. The model incorporated the mathematical relationships and input parameters necessary to calculate the hydraulic and electrical performance indicators of each configuration. The simulation results were then compared with theoretical values obtained from established engineering equations and design specifications.

The comparison aimed to determine whether the simulated outputs were consistent with the theoretical calculations. The evaluated parameters included volumetric flow rate, nozzle discharge velocity, water consumption, electrical power demand, and energy consumption,

depending on the available model outputs and the applicable design configuration. The proponents examined the differences between the theoretical and simulated results to identify possible inconsistencies in the mathematical model, input parameters, and computational methods.

Differences between theoretical and simulated values may arise from variations in the assumptions, equations, input specifications, and computational methods used in the analysis. For example, theoretical calculations may assume steady flow or ideal operating conditions, whereas a more detailed model may incorporate additional system characteristics. Similarly, variations in the assumed pump efficiency, operating pressure, nozzle dimensions, or operating duration may influence the calculated results.

The proponents used these comparisons to assess the consistency of the simulation model and determine whether the outputs reasonably represented the expected behavior of each proposed design. Any numerical discrepancies should be quantified using the actual results obtained from the theoretical calculations and MATLAB Simulink simulations. The percentage error may be determined using the theoretical value as the reference, provided that the reference value is nonzero.

Through this process, the proponents established a basis for evaluating the reliability of the simulation results and identifying the parameters that required further verification. The findings also supported the comparative assessment of the three configurations in terms of water utilization, hydraulic performance, electrical energy requirements, and operational feasibility.

3. Comparative Evaluation of the Proposed Designs

The proponents compared the three proposed designs using the performance parameters established for the study. These parameters provided a structured basis for identifying the advantages and limitations of each configuration. The comparison considered hydraulic performance, water consumption, electrical energy requirements, cleaning coverage, operational characteristics, and estimated implementation cost, subject to the availability of the corresponding calculations and simulation results.

Design 1 relied primarily on the wall mounted sprinkler mechanism to distribute water across the pigpen area. Its performance depended on the arrangement of the spray nozzles, water pressure, flow rate, and operating duration. The evaluation of this design therefore focused on its ability to distribute water effectively while maintaining acceptable water consumption and electrical energy requirements.

Design 2 incorporated modifications to the wall mounted sprinkler configuration. The proponents evaluated the effects of these modifications on the relevant hydraulic parameters and expected cleaning coverage. The comparative analysis considered whether the revised arrangement improved the distribution of water or reduced resource requirements relative to the first design. These improvements must be established through the actual calculations and simulation results rather than inferred from the design modifications alone.

Design 3 integrated a mechanical scraping mechanism with a nozzle based washing system. This configuration introduced an additional cleaning mechanism that could assist in removing accumulated solid waste before or during the washing process, depending on the operating sequence. The analysis therefore considered the combined electrical requirements of the scraping and water delivery mechanisms, together with the hydraulic performance and estimated operating time of the system.

The proponents also considered the estimated implementation cost of each configuration. The bill of materials provided the basis for determining the financial requirements associated with the components and materials necessary for system construction. A lower estimated cost may offer an economic advantage, but the final selection should also account for cleaning performance, water consumption, electrical energy requirements, reliability, and maintenance considerations.

The comparative evaluation enabled the proponents to examine the tradeoffs among the three configurations. Rather than selecting a design based on a single parameter, the study considered the combined results of the engineering calculations, simulation outputs, and cost estimates. The most suitable design should demonstrate an appropriate balance between cleaning effectiveness, resource utilization, electrical safety, operational reliability, and implementation cost.

5.2. Summary of Findings

The study focused on the design, simulation, and analysis of an automated pigpen washing system through the development and evaluation of three proposed configurations. The proponents examined the mechanical, hydraulic, and electrical requirements of each design and used mathematical calculations and MATLAB Simulink simulations to evaluate their operational characteristics.

1. **Design and development of the automated pigpen washing system.** The proponents developed three design configurations consisting of a wall mounted sprinkler washing system, an improved wall mounted sprinkler washing system, and a scraper and nozzle combination washing system. Each configuration incorporated a distinct arrangement of mechanical, hydraulic, and electrical components to support the automated cleaning

process. The proponents prepared the design layouts, circuit diagrams, and bills of materials to establish the technical and material requirements of the proposed systems.

2. **Hydraulic and electrical performance evaluation.** The proponents evaluated the hydraulic characteristics of the proposed designs using relevant fluid mechanics equations and design parameters. The analysis considered nozzle cross sectional area, volumetric flow rate, discharge velocity, and Reynolds number, where applicable. The proponents also examined the electrical power and energy requirements associated with the operation of the system components. These calculations provided the basis for assessing the water delivery characteristics, resource requirements, and electrical demands of the proposed configurations.
3. **Simulation and comparison of theoretical results.** The proponents developed MATLAB Simulink models to calculate and examine the operational parameters of the three proposed designs. The simulated outputs were compared with theoretical values derived from established engineering equations. This process supported the assessment of computational consistency and helped identify differences that could result from variations in assumptions, input parameters, and calculation methods. The final assessment of simulation accuracy requires the corresponding numerical results and percentage errors for each evaluated parameter.
4. **Material requirements and estimated implementation cost.** The proponents prepared separate bills of materials for the three proposed designs. Each bill of materials identified the required components, quantities, unit prices, and estimated total costs. This documentation provided a basis for comparing the material requirements and financial implications of each configuration. The actual cost comparison should use the finalized quantities and verified prices of the components included in each design.
5. **Comparative assessment of the proposed configurations.** The proponents established a framework for comparing the three designs in terms of hydraulic

performance, water consumption, electrical energy requirements, cleaning coverage, operational characteristics, and estimated implementation cost. Design 1 emphasized water distribution through a wall mounted sprinkler mechanism, while Design 2 incorporated modifications to improve the sprinkler arrangement. Design 3 combined mechanical scraping with water spraying to support solid waste removal and subsequent washing. The final ranking of the designs depends on the actual simulation results, engineering calculations, and cost estimates obtained during the study.

Overall, the findings established the design and analytical framework for evaluating the proposed automated pigpen washing system. The combination of engineering calculations, simulation, and cost estimation enabled the proponents to assess the technical feasibility of the three configurations and identify the parameters that should guide the selection of the most suitable design.

5.3. Conclusion

Based on the objectives and analytical procedures of the study, the following conclusions are established:

1. The study developed three proposed automated pigpen washing system configurations that applied different cleaning mechanisms to address the removal of accumulated waste and the distribution of water within the pigpen area. The proponents prepared the corresponding design layouts, circuit diagrams, and bills of materials to define the mechanical, hydraulic, and electrical requirements of each configuration. These outputs established the technical foundation for the subsequent simulation and comparative analysis.

2. The study applied engineering equations and MATLAB Simulink modeling to evaluate the hydraulic and electrical characteristics of the proposed designs. The analysis considered relevant parameters such as volumetric flow rate, nozzle discharge velocity, water consumption, electrical power demand, and energy consumption. Comparing theoretical and simulated results provided a method for examining the consistency of the mathematical models and identifying potential differences in the calculated outputs. However, the validity and accuracy of each simulation must be established through the actual numerical results and, where feasible, experimental verification.
3. The study established a comparative evaluation framework for assessing the three proposed configurations. The framework considered cleaning performance, hydraulic requirements, water consumption, electrical energy requirements, operational characteristics, and estimated implementation cost. The results of this comparison can support the identification of a design that satisfies the intended operational requirements while maintaining appropriate resource utilization and electrical safety.
4. The integration of mechanical, hydraulic, and electrical design considerations demonstrated the importance of evaluating the washing system as a coordinated engineering system rather than as a collection of independent components. The effectiveness of each configuration depends on the compatibility of its components, the adequacy of its control mechanisms, and its ability to meet the required cleaning performance under the specified operating conditions.

The final determination of the most suitable design should rely on the verified simulation results, theoretical calculations, finalized cost estimates, and any experimental findings obtained by the proponents. These results will establish whether the selected configuration provides the most appropriate balance between cleaning effectiveness, water and energy efficiency, operational reliability, and implementation cost.

5.4. Recommendations

Based on the scope and methodology of the study, the following recommendations are proposed to support further development, validation, and improvement of the automated pigpen washing system.

1. **Conduct prototype testing under actual operating conditions.** Future researchers should construct and test a prototype of the selected design to verify the hydraulic, mechanical, and electrical performance predicted by the simulations. Experimental measurements should include water flow rate, operating pressure, water consumption, electrical power consumption, and cleaning time. These measurements should be compared with the theoretical and simulated values to determine the accuracy of the mathematical models and identify areas requiring modification.
2. **Improve and validate the MATLAB Simulink models.** Future researchers should refine the simulation models by incorporating more representative operating parameters, including pump efficiency, pressure losses, nozzle discharge characteristics, motor efficiency, and variations in operating conditions where applicable. Researchers should also verify the equations, input values, and signal connections used in each subsystem. The validated models can then support more reliable comparisons among the proposed designs and facilitate the evaluation of additional operating scenarios.
3. **Evaluate water and electrical energy consumption.** Future studies should measure the water and electrical energy consumed during a complete washing cycle. Researchers should determine the relationship between operating duration, water flow rate, cleaning coverage, and energy requirements. These measurements can help identify opportunities to reduce resource consumption without compromising the effectiveness of the washing process.

4. **Assess cleaning effectiveness and operational reliability.** Future researchers should evaluate the ability of each design to remove accumulated waste from different areas of the pigpen. The assessment should consider cleaning coverage, residual waste, cycle duration, and the consistency of the washing process. For the scraper and nozzle combination design, researchers should also evaluate the mechanical performance of the scraping mechanism and its coordination with the water delivery system.
5. **Enhance the electrical control and protection systems.** Future development should incorporate suitable electrical protection devices, control components, and sensor arrangements based on the system's operating requirements. Researchers should assess the reliability of the automated control sequence and verify that the electrical components are appropriately protected against moisture and other environmental conditions associated with pigpen operation. The final installation should comply with applicable electrical safety requirements.
6. **Conduct a comprehensive cost and maintenance assessment.** Future researchers should update the bill of materials using verified supplier quotations and actual component requirements. The evaluation should include initial implementation cost, water expenses, electrical energy costs, replacement components, and routine maintenance. This assessment would provide a more complete basis for determining the long term economic feasibility of the selected design.
7. **Optimize the system configuration through further design studies.** Future researchers should investigate alternative nozzle arrangements, operating pressures, scraping mechanisms, and control sequences to improve cleaning performance and resource efficiency. Additional simulations can be conducted to determine how changes in design parameters affect water consumption, electrical energy requirements, and cleaning coverage. The results may support further optimization of the system before full scale implementation.

8. **Conduct extended testing in an actual pigpen environment.** Researchers should evaluate the selected design over repeated washing cycles and under different operating conditions. Extended testing can help identify potential issues involving component wear, nozzle blockage, corrosion, moisture exposure, and control system reliability. The resulting observations can guide improvements to the design and establish more reliable maintenance procedures before practical deployment.